

Energy Education and Research Initiatives in India

Renewable energy sector is very important for a country like India for a faster growth. Realising its importance, the government of India and others working in this field have already taken some initiatives, which need to be highlighted. Here are some of them



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In 21st century energy has become a major concern in India. We are in dire need of alternate energy (renewable) sources like solar, wind, tidal and their combinations. Due to this serious concern, we need to educate our young generation about energy consciousness, energy research and energy education at various levels. Fortunately, realising this, the government of India and some others have already taken some initiatives in this direction. Here are some of them:

1. The government of India initiated solar energy research initiative (SERI) under Department of Science and Technology (DST) to facilitate and promote R&D activities in solar based

research in India in association with industries.

2. Ministry of New & Renewable Energy (MNRE) of the government extended its funding for energy research (solar, wind, tidal, hydrogen and hybrid) to academic institutes and research labs.

3. The Department of Science & Technology (DST) also recently encouraged industries and R&D centres to establish Technology, Information, Forecasting & Assessment Council - Centre of Relevance & Excellence (TIFAC-CORE) in the area of SPV technology.

4. MNRE introduced National Renewable Energy Fellowship (NREF) scheme for JRF/SRF/RA and post-doctoral Fellows working in the area of renewable energy.

5. Waste to energy scheme of the government is encouraging energy resources.

6. Jawaharlal Nehru National Solar Mission (JNNSM) set up by the government to enhance usage of solar energy.

7. Establishment of solar cell hub by DST-Technology Mission at IIST (formerly BESU), Kolkata.

8. Establishment of advanced centre for photo-voltaic by MNRE at IIST, Kolkata.

9. National Centre for Photovoltaic Research and Education (NCPRE) at IIT Bombay was launched in 2010 as part of the JNNSM.

10. Centre of excellence on Renewable Energy Systems at NIT, Rourkela (Odisha) funded by NPIU under TE-QIP, MHRD, New Delhi.

Besides, following premier institutions in India are offering courses in energy science, technology, engineering and management at undergraduate, postgraduate and doctoral levels (see table).

The Energy and Resources Institute, commonly known as TERI (formerly Tata Energy Research Institute), established in 1974, is a research institute based in New Delhi focusing its research activities in the fields of energy, environment and sustainable development. According to Think Tanks and Civil Societies Program (TTCSP) at the International Relations Program, University of Pennsylvania, TERI was ranked 20 in the list of top global think tanks on environment.

There are also some institutes under the MNRE, such as: